Specialized test procedure—Prover shell temperature corrections

Category: Volume
Part: 4-STP
Section: 23
Publication date: 2019-12-19
Revision number: 3

Application

This procedure is used to correct all open neck prover shell volumes back to the appropriate reference temperature when performing metering system accuracy tests using these provers. While the calculations are similar for closed-circuit provers such as pipe provers and compact provers, they are not intended to be addressed by this bulletin. Please contact your volumetric specialist for more information on using closed-circuit provers.

Note: Typically, the appropriate reference temperature is 15 °C except for sanitary provers used to test milk metering systems, which have a reference temperature of 4.4 °C, and for glassware, which has a reference temperature of 20 °C.

Purpose

Provers are calibrated to contain their indicated nominal capacity when the prover shell temperature is at the reference temperature, which is usually 15 °C. If the temperature of the prover shell differs from the reference temperature, volume correction for the expansion or contraction of the prover shell will need to be calculated into the observed volume reading.

Note: In practical terms, the product temperature in the prover can be assumed to be the same as the prover shell temperature. This allows for the use of a thermowell mounted in the prover body to be used for determining prover shell temperatures. Special consideration should be made to ensure that the prover shell is not subject to localized temperature disturbances which may affect the temperature readings.

Reference

Bulletin V-17—Temperature Correction of Provers

Note: Bulletin V-17 provides a table of correction factors in relation to the prover size, temperature and material from which the prover is made.

Procedure

The following formula will derive the exact correction required:

$$\mathrm{Ctsp} = \left(\mathrm{T_p} - \mathrm{T_{REF}}\right) \times CCE \times Vref$$

Where:

T_p = Temperature of the prover shell

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T_{REF} = Reference temperature (usually 15 °C for fuel provers and 4.4 °C for milk provers)

CCE = Cubical coefficient of thermal expansion per degree Celsius for the prover material

Ctsp = Volume correction for the temperature of the steel in the prover shell

Vref = the certified base volume of the prover at the reference temperature

Note: The CCE is often found on the prover's nameplate. Always use the indicated CCE if available. In some cases, it may be identified with Greek letters, typically Alpha (α) or Gamma (γ). Here are some CCEs for common prover construction materials:

304 stainless steel = 5.18×10^{-5} or 0.000 0518 / °C

316 stainless steel = 4.54×10^{-5} or 0.000 0454 / °C

17.4 stainless steel = 3.24×10^{-5} or 0.000 0324 / °C

Mild steel = 3.35×10^{-5} or 0.000 0335 / °C

Borosilicate glassware (Pyrex®, Bomex®, Duran®, etc.) = 1.00×10^{-5} or 0.000 010 / °C

Note: Please see the Guidelines for the Calibration and Certification of Volumetric Standards for further information and other CCE values.

Example

A thermometer installed in a 500 L 304 stainless steel prover shell indicates 6 °C and the uncorrected volume indicated by the prover is 499.9 L. The prover is a fuel prover designed to be used at a reference temperature of 15 °C.

Correction of the observed prover reading for the contraction of the prover shell is necessary.

Volume correction = 500 L x 0.0000518 x (6 °C - 15 °C) = -0.233 L or -233 ml

Corrected volume = 499.9 L + (-0.233 L)

= 499.667 L

≈ 499.670 L (rounded off to prover resolution)

Therefore, after filling the prover and reading the volume indicated at the end of a proving run, you must subtract 233 ml (add - 233 mL) to the reading. In other words, the prover actually contains 233 ml less than it indicates, at 6 °C.

Note: Rounding may be applied for calculated correction values less than the resolution of the prover's gauge plate.

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Revisions

Revision 3

The purpose of revision 3 was to:

- remove references to the deprecated value for Seraphin® stainless steel;
- clarify that shell corrections should be applied to all provers regardless of size;
- clarify the scope to ensure the calculations are only used for the correction of open neck provers and measures (pipe provers and compact pipe provers have special considerations not currently covered by the bulletin);
- correct the CCE for borosilicate glassware to show the cubical coefficient. (Previously, the linear coefficient was mistakenly shown. In all cases, the actual CCE for the shell being examined should be used.)

Revision 2

The purpose of revision 2 was to:

- correct the sign of the volume calculation result;
- update the temperature format for consistency with other documents;
- change applicability to all open provers by including the concept of reference temperature;
- add cubical coefficient of thermal expansion for Seraphin® stainless steel and borosilicate glass (Pyrex®, etc.);
- standardized scientific notation format.

Revision 1

The purpose of revision 1 was to correct a typographical error for the reference temperature of a milk prover. Milk provers are calibrated to contain their nominal volume at a reference temperature of 4.4 °C, which is the temperature of the milk at the time of inspection.